

UrbanFix: A Closed-Loop, AI-Validated Urban Governance Platform with a Public Accountability Registry and Privacy-Preserving Research Data Access

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Abstract—Most digital grievance systems classify the text of a complaint and track it as a ticket. The photograph a citizen attaches is rarely checked, urgency comes from an opaque model if it is computed at all, and the records are seen only by the administration. We present UrbanFix, an urban governance platform that follows a complaint from the citizen’s photograph to a verified, publicly recorded repair and makes the resulting data available for research. Citizens file without an account, confirming each report with a one-time email code. A YOLOv8 model trained for the selected category checks and localises the defect in each photo, and a weighted questionnaire produces an urgency score that can be recomputed from the answers. Staff work through a seven-stage workflow enforced on the server, which requires assignment within the ward, photo proof and a resolution receipt before a complaint is closed. All complaints and their histories appear in an open registry without personal data, and approved researchers get time-limited, logged access to an anonymised projection. We offer only categories for which a public annotated dataset exists, so every accepted report can be checked by a model. On the validation sets, mask mAP50 is 0.724 for potholes and 0.502 for litter, and box mAP50 is 0.908 for open manholes and 0.729 for graffiti, with 19 to 33 ms per image on a laptop CPU. When the model is unsure, it flags the report for a human reviewer.

Index Terms—e-governance, smart cities, civic technology, object detection, YOLOv8, human-in-the-loop AI, open data, privacy-preserving data sharing

I. INTRODUCTION

In India, pothole-related road accidents rose from 3,713 in 2020 to 5,432 in 2024, an increase of 46%, and pothole-related accidents killed 8,963 people between 2018 and 2022 [1], [2]. The national grievance portal CPGRAMS received between 18.7 and 26.2 lakh grievances a year from 2019 to 2024 [3], [4], and in FY 2023–24 State and Union Territory governments, which handle municipal complaints, took 119 days on average to dispose of one, against 16 days for central ministries [5] (Fig. 1).

Recent systems automate parts of this process. Some route complaint text [7], [12], one predicts urgency [8], and others classify waste images [9] or filter unsafe photos [10]. None

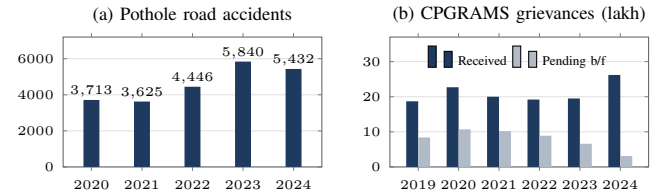


Fig. 1. Motivation: (a) pothole-related road accidents in India [2]; (b) grievances received on CPGRAMS and the backlog brought forward each year [3], [4].

of these complaint management tools checks whether a photo shows the reported defect. Their urgency estimates cannot be traced back to an input, and their records are visible only to officials or to the person who filed. Holding a municipality to account also needs a record of who handled each step, a public outcome, and data that others can study.

UrbanFix handles a complaint as a governance record that passes through six steps in one system: report, validate, prioritise, act, prove and learn. This paper makes the following contributions:

- 1) A vision check for each category, offered only when a public, annotated street-level dataset exists for it; a YOLOv8 model [18] trained on that dataset localises the defect in each photo.
- 2) A prioritisation scheme that keeps the citizen’s weighted questionnaire score and the model’s evidence as separate signals and leaves the decision to a supervisor; the model cannot reject a report.
- 3) A workflow modelled as a guarded finite-state machine, with assignment limited to the ward, required photo proof, a rework path and an append-only history.
- 4) A public registry that lists every complaint with its history; the server’s queries remove the citizen’s identity before anything is returned.
- 5) A research data gateway that gives approved researchers

TABLE I
CAPABILITIES OF REVIEWED SYSTEMS AND URBANFIX (UF)

Capability	[6]	[7]	[8]	[9]	[10]	[11]	[12]	UF
Account-free filing	○	●	○	○	○	○	○	●
Photo evidence	●	○	●	●	●	●	●	●
Photo-defect validation	○	○	●	○	○	○	○	●
Urgency scoring	○	○	●	○	○	●	●	●
Traceable urgency	○	○	○	○	○	○	○	●
Automated routing	●	●	●	●	●	●	●	●
Field proof of repair	●	○	○	○	○	○	○	●
Resolution receipt	●	○	○	○	○	○	○	●
Public registry	○	○	○	○	○	○	○	●
Open research data	○	○	○	○	○	○	○	●

● supported ● partial ○ not supported or not reported

an eight-field pseudonymised view of the data, with a granted scope, an expiry date, daily export limits and an access log.

Section II covers related work and Section III the architecture. Section IV describes the methods, Section V the implementation and Section VI the results; Section VII concludes.

II. RELATED WORK

Infra-Alert [6] collects geotagged photos and produces PDF reports but leaves verification and prioritisation to future work. CitiWatch [7] routes complaint text with TF-IDF and a supervised classifier at 96.8% accuracy. The framework of Cholke *et al.* [8] combines Sentence-BERT features, a multi-task urgency network and DBSCAN duplicate grouping, reporting 81% faster response. ReGenX [9] classifies waste photos on device (85%), CivicMate [10] screens photos for unsafe content and lets citizens upvote local issues, CivicX [11] pairs grievance analytics with plan approval and benchmarks its classifier against officers (91.4% vs. 94.1%), and Smart City Concern [12] uses zero-shot BART with sentiment as a proxy for urgency.

Reported accuracy for text routing lies between 90 and 97%. As Table I shows, none of the seven checks the photo against the reported defect, gives an urgency that can be traced to its inputs, publishes all complaints with their histories, or shares data with researchers. Datasets with box or mask labels now exist for road damage [13], [14], litter [15], open manholes [16] and graffiti [17], which makes defect-level checks feasible for these categories.

III. SYSTEM ARCHITECTURE

Fig. 2 shows the layers. One Expo (React Native) client serves all six roles, and the server decides which set of screens a user sees. Each request passes through an Express pipeline before reaching one of seven domain services: transport checks, upload handling, JWT authentication, a per-route role guard and schema validation. The authentication step also re-checks on every call whether the account is active, whether an invitation is still pending and whether a researcher’s access has expired. The intake service sends photos to a separate FastAPI inference service only after the tracking ID has been returned,

so a slow or failed model call does not affect the filing. MongoDB holds seven collections. Each complaint document embeds its status history and its AI analysis.

IV. METHODOLOGY

A. Category selection by dataset availability

Let $\mathcal{K}$ be the candidate complaint categories. A category k is offered only if a public dataset $\mathcal{D}_k$ of street-level images with localised labels (boxes or masks), documented provenance and a usable licence exists:

$$\mathcal{C} = \{k \in \mathcal{K} : \mathcal{D}_k \text{ is public} \wedge \text{localised} \wedge \text{street-level}\}. \quad (1)$$

Four of the ten categories we started with meet this condition: pothole/road damage, garbage/litter, open manhole and graffiti. For water leakage, streetlights, illegal parking, fallen trees, road signs and electric poles we found only small collections, image-level labels or data without a licence, so these were left out. The API rejects any category outside $\mathcal{C}$.

B. Traceable questionnaire urgency

Each category has five yes/no questions q_i with weights $w_i \in \{1, 2\}$ (2 for safety-critical). With answers $y_i \in \{0, 1\}$, the urgency score and band are

$$U = \frac{\sum_{i=1}^5 w_i y_i}{\sum_{i=1}^5 w_i}, \quad u_Q = \begin{cases} \text{High,} & U \geq 0.6 \\ \text{Medium,} & 0.3 \leq U < 0.6 \\ \text{Standard,} & U < 0.3. \end{cases} \quad (2)$$

Dividing by $\sum w_i$ makes scores comparable across categories whose questions carry different weights ($\sum w_i = 7$ for pothole and manhole, 6 for litter and graffiti). For a pothole, one safety-critical “yes” gives $U = 0.29$, which is Standard, and two give 0.57, which is Medium. The citizen sees u_Q while filing, and the stored answers are enough to recompute it (Fig. 3).

C. Per-category vision validation

For a complaint of category k with photos $x_1, \dots, x_n$ ($n \leq 3$), the model f_k returns detections $\mathcal{B}_m = f_k(x_m) = \{(b_j, s_j, M_j)\}$ with boxes b_j , confidences s_j and, for segmentation models, masks M_j . The service keeps detections with $s_j \geq \tau$ ($\tau = 0.25$) after non-maximum suppression and computes

$$s^* = \max_m \max_j s_j, \quad (3)$$

$$\delta = \max_m \frac{|\bigcup_j M_j^{(m)}|}{H_m W_m}, \quad (4)$$

$$r = \mathbf{1}[s^* < \tau], \quad (5)$$

where δ is the share of the image covered by the predicted masks and r is the `needsReview` flag. The suggested urgency u_{AI} equals u_Q , raised by one band when δ is above a configurable threshold θ_δ . The model does not change the stage of a complaint. The supervisor sees u_Q , u_{AI} , r and the annotated photos together and makes the decision (Fig. 4). If the model misses a defect, the result is extra review work for staff; the report itself is kept.

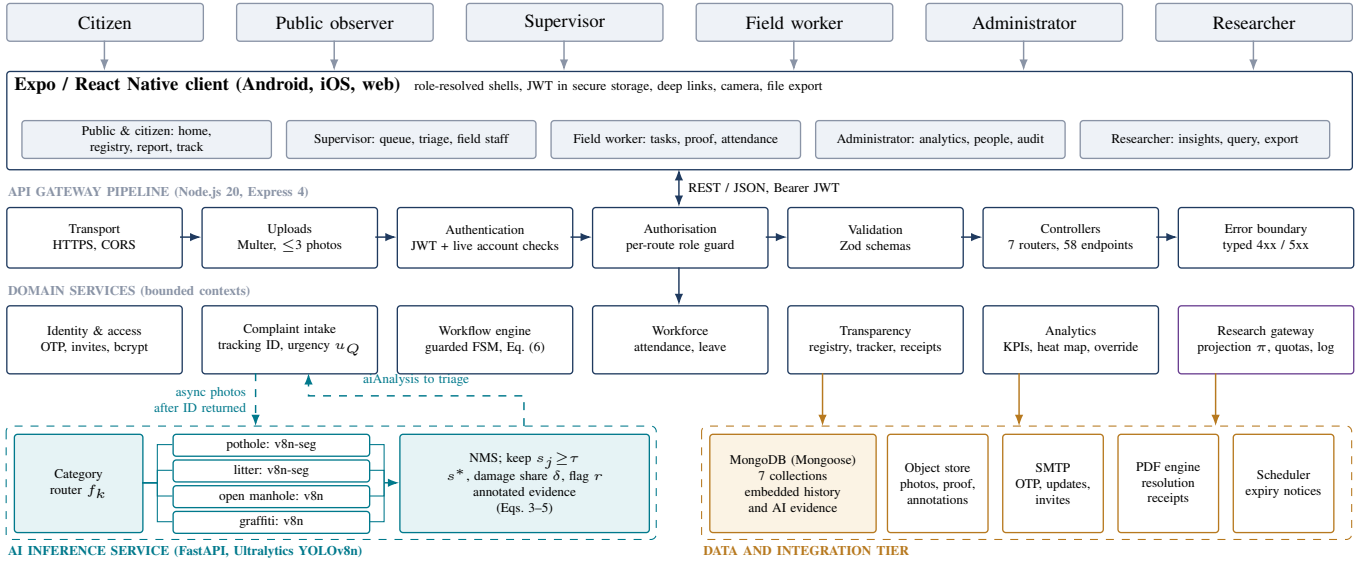


Fig. 2. Layered architecture of UrbanFix. Solid arrows are synchronous requests. Dashed arrows show the AI call, which is made asynchronously after the tracking ID has been returned. The security and privacy controls of Sections IV-D–IV-E apply to all layers.

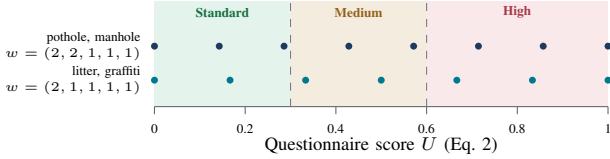


Fig. 3. All reachable questionnaire scores for the two weight profiles, with the urgency bands. For a pothole, a single safety-critical “yes” scores $2/7 = 0.29$ and stays in Standard.

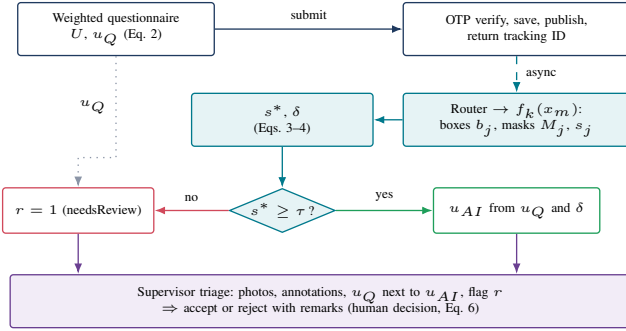


Fig. 4. AI validation and prioritisation. The supervisor sees both urgency signals; a low-confidence result sets *needsReview* and the report stays in the queue.

D. Accountable workflow

The workflow is a deterministic finite-state machine $\mathcal{M} = (S, A, \Delta, s_0, F)$ (Fig. 6) with

$$S = \{\text{sub}, \text{acc}, \text{asg}, \text{wip}, \text{prf}, \text{cls}, \text{rej}\},$$

$$A = \{\text{accept}, \text{reject}, \text{assign}, \text{start}, \text{proof}, \text{close}, \text{rework}\},$$

start $s_0 = \text{sub}$ and terminal set $F = \{\text{cls}, \text{rej}\}$. A transition is applied only if its guard holds:

$$\Delta(s, a) \text{ is defined } \iff s \in \mathcal{S}_a \wedge \rho(\text{actor}) \in \mathcal{R}_a \wedge g_a(c), \quad (6)$$

where $\mathcal{S}_a$ and $\mathcal{R}_a$ are the source stages and roles allowed for action a , and g_a requires remarks of at least 10 characters for decisions, a field worker in the complaint’s ward who is active and not on approved leave for *assign*, the assignee only for *start* and *proof*, and up to three proof photos for *proof*. Each transition that is applied adds (stage, status, actor, remarks, t) to the history, which is never edited, and *close* emails a PDF receipt to the citizen. The public sees a coarser status given by $\varphi : S \rightarrow \{\text{Pending}, \text{In Progress}, \text{Resolved}, \text{Rejected}\}$ with $\varphi(\text{sub}) = \text{Pending}$, $\varphi(\text{cls}) = \text{Resolved}$, $\varphi(\text{rej}) = \text{Rejected}$ and *In Progress* otherwise.

E. Tiered disclosure and research projection

Different audiences receive different server-side projections of the same record (Table II). The registry query leaves out the citizen and staff references, and the tracker shows the filer in the history only as “Citizen”. Researchers receive the eight-field projection

$$\pi(c) = (h_\kappa(c.id), k, u_Q, \varphi, \text{ward}, \lfloor t_f \rfloor_d, \lfloor t_c \rfloor_d, \Delta t), \quad (7)$$

$$h_\kappa(x) = \text{trunc}_{48}(\text{HMAC-SHA256}(\kappa, x)), \quad \Delta t = \frac{t_c - t_f}{86400 \text{ s}}, \quad (8)$$

where $\lfloor \cdot \rfloor_d$ truncates a timestamp to the day and κ is a server secret. The record ID stays the same across exports, so researchers can join them, but it cannot be mapped back to the database or tracking ID without κ . Identity, free text, street address, tracking ID and photos are not included. Access is granted per application with scope $\sigma \in \{\text{aggregate}, \text{records}\}$

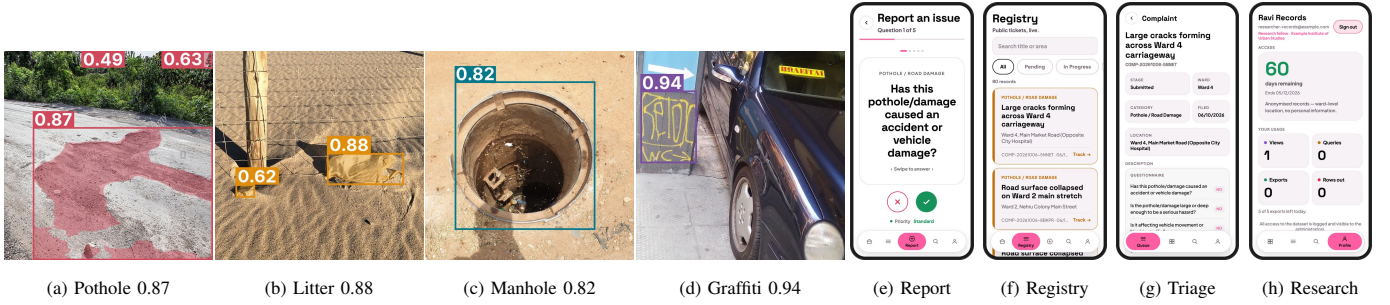


Fig. 7. (a)–(d) Stored AI evidence on validation images: predicted masks or boxes with confidence. (e)–(h) App screens: weighted swipe questionnaire with live urgency, open public registry, supervisor triage, and researcher access with expiry countdown, quota and audit notice.

TABLE III
DATASETS AND MODELS PER CATEGORY

Category	Dataset	Labels	Train/val	Model
Pothole / road damage	Roboflow [14]	masks	720/60	v8n-seg
Garbage / litter	TACO [15]	masks [†]	1214/214	v8n-seg
Open manhole	Road Hazards [16]	boxes	1123/480	v8n
Graffiti	STORM [17]	boxes	813/209	v8n

[†]60 TACO classes merged into one; 1,428 of 1,500 images still available.

TABLE IV
VALIDATION RESULTS (BEST CHECKPOINT)

Model (metric set)	P	R	mAP50	mAP50–95	CPU ms
Pothole (mask)	0.725	0.711	0.724	0.426	33
Pothole (box)	0.733	0.682	0.719	0.447	
Litter (mask)	0.757	0.434	0.502	0.317	24
Litter (box)	0.748	0.446	0.521	0.375	
Open manhole (box)	0.890	0.874	0.908	0.469	19
Graffiti (box)	0.836	0.604	0.729	0.526	20

0.908), probably helped by the regular shape of manholes, although the dataset contains augmented copies of some images that may raise this figure. The pothole model reaches a mask mAP50 of 0.724 with similar precision and recall. The graffiti model has high precision (0.836) but misses faint tags (recall 0.604). Litter is the hardest category (mask mAP50 0.502): TACO’s 60 object classes were merged into one and many objects are small. Given these recall values, rejecting reports on the model’s say would lose real complaints, which is why Eq. (5) only flags them. Inference takes 19 to 33 ms per photo on the laptop CPU and 6 to 20 ms on its GPU (Fig. 10), so a GPU server is not required. Fig. 7 shows stored outputs and the matching app screens.

B. System validation

We ran sixteen API tests against a live server with a separate database, and all of them passed. They cover OTP validation and expiry, rejection of categories outside $\mathcal{C}$, the absence of citizen identity in registry and tracker responses, role checks (a field worker trying to accept gets 403), the 10-character remarks rule, a *reject* after *accept*, assignment across wards, a receipt request before closure, and an aggregate-

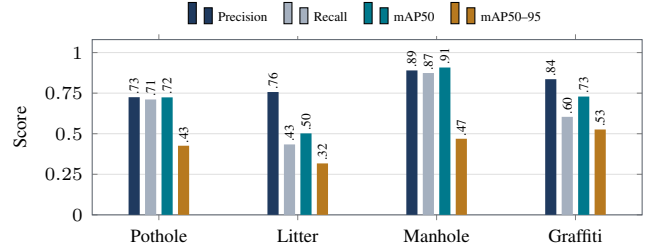


Fig. 8. Validation metrics per model (mask metrics for the segmentation models, box metrics for the detection models).

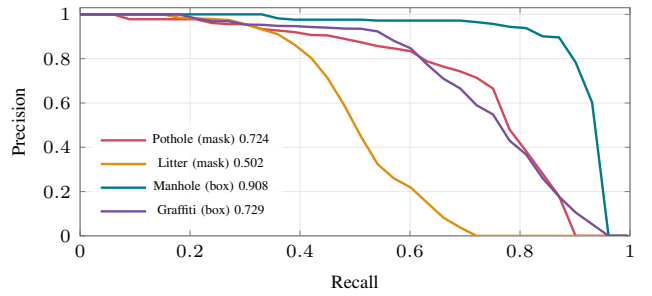


Fig. 9. Precision–recall curves on the validation sets; legend gives AP at IoU 0.5.

only researcher asking for records (403). Each test targets one condition in Eqs. (6) or (9).

C. Workflow on a demonstration dataset

To test the dashboards, we seeded 80 complaints spread over twelve months and moved them through the stages (Fig. 11). Of these, 41 were accepted and 9 rejected at triage, and 19 were closed with a receipt, taking 8.2 to 9.2 days on average depending on the category. Because the data are synthetic, these numbers only demonstrate the pipeline and dashboards.

D. Limitations

Only four categories have suitable public data, so common complaints such as water leakage and faulty streetlights are not supported yet. The training images come from other countries, and we have not measured accuracy on Indian street photos. We chose the questionnaire weights and θ_δ ourselves instead of fitting them to outcomes. Location is stored as a ward and

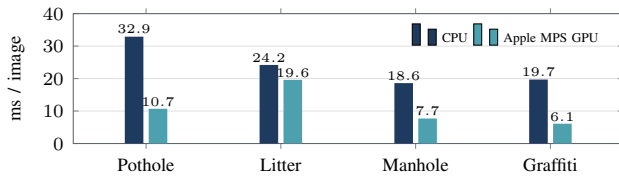


Fig. 10. Mean inference time per 640px image including pre- and post-processing (40 images, batch 1, after warm-up).

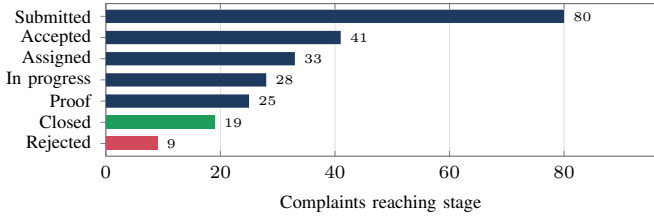


Fig. 11. Complaints reaching each stage on the seeded demonstration dataset ($n = 80$).

free text, with no GPS and no duplicate detection. Finally, the workflow figures come from seeded data, not a field pilot.

VII. CONCLUSION AND FUTURE WORK

UrbanFix handles a civic complaint from the citizen's photo to a recorded repair using open-source software and public datasets. Each photo is checked by a model trained for its category, a supervisor sets the priority with both the citizen's score and the model's evidence in view, and the server enforces each workflow step, including photo proof before closure. Records are published without personal data, and researchers get logged, pseudonymised access. Next, we plan to fine-tune the models on locally collected images, fit the questionnaire weights to closed complaints, add GPS and duplicate detection, and support regional languages and SMS reporting. We also plan a ward-level pilot with a municipal corporation to compare the model's checks with manual review.

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